

Assignment 7.

1. We know z-gate is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

When we apply z gate to $|-\rangle$

$$Z|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle$$

2. We know control phase shift gate is

$$U_{CPS, \phi} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\phi} \end{bmatrix}$$

In z-axis when angle is π we can assume

$$e^{i\phi} = \cos\phi + i\sin\phi$$

$$e^{i\pi} = \cos\pi + i\sin\pi = -1$$

$$\text{Controlled z gate is } \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$\text{Since } Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow Z|0\rangle = |0\rangle, Z|1\rangle = -|1\rangle$$

When controlled z gate applied to $|+\rangle$

$$Z|+\rangle = Z \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) = \frac{Z|0\rangle + Z|1\rangle}{\sqrt{2}} = \frac{|0\rangle + (-|1\rangle)}{\sqrt{2}} = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

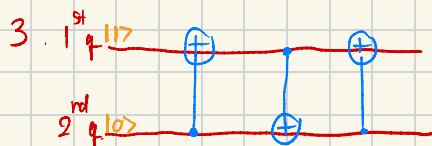
$$Z|-\rangle = Z \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = \frac{Z|0\rangle - Z|1\rangle}{\sqrt{2}} = \frac{|0\rangle - (-|1\rangle)}{\sqrt{2}} = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$

In the $| \pm \rangle$ basis, the z gate acts exactly like a classical NOT (or Pauli-X) gate does in the computational basis. So, if we wrap the target qubit of a CZ gate in H gates, we transition the target qubit into the $| \pm \rangle$ basis, apply the conditional Z operation (which acts as a NOT in this basis), and transition back. This makes the entire operation equivalent to a CNOT gate.

$$I \otimes H = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

$$CZ(I \otimes H) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

$$(I \otimes H)[CZ(I \otimes H)] = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \text{CNOT gate}$$



1st CNOT gate: $CNOT_1(|00\rangle = |00\rangle, |01\rangle = |01\rangle, |10\rangle = |11\rangle, |11\rangle = |10\rangle)$

2nd CNOT gate: $CNOT_2(|00\rangle = |00\rangle, |01\rangle = |11\rangle, |11\rangle = |01\rangle, |10\rangle = |10\rangle)$

3rd CNOT gate: $CNOT_3(|00\rangle = |00\rangle, |11\rangle = |10\rangle, |01\rangle = |01\rangle, |10\rangle = |11\rangle)$

$$CNOT_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix}$$

4. Given $H|\psi\rangle = \alpha|+\rangle + \beta|-\rangle$

Normally we can directly understand that coefficient of $|0\rangle$ & $|1\rangle$ basis is

$$\frac{1}{\sqrt{2}}(\alpha + \beta) \text{ \& \> } \frac{1}{\sqrt{2}}(\alpha - \beta)$$

We can derive from

$$= \alpha|+\rangle + \beta|-\rangle$$

We now $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

$$= \alpha\left(\frac{1}{\sqrt{2}}|0\rangle + |1\rangle\right) + \beta\left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right)$$

$$= \frac{\alpha}{\sqrt{2}}|0\rangle + \frac{\alpha}{\sqrt{2}}|1\rangle + \frac{\beta}{\sqrt{2}}|0\rangle - \frac{\beta}{\sqrt{2}}|1\rangle$$

$$H|\psi\rangle = \frac{1}{\sqrt{2}}(\alpha + \beta)|0\rangle + \frac{1}{\sqrt{2}}(\alpha - \beta)|1\rangle$$

5. $H^{\otimes n}|\psi\rangle = \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} (-1)^{x \cdot y} |x\rangle$

Here n means n qubit Hadamard gate.

$$H^{\otimes 2}|2\rangle = \frac{1}{2^{2/2}} \sum_{x=0}^{2^2-1} (-1)^{x \cdot 2}$$

$$= \frac{1}{2} \left((-1)^{0 \cdot 2} |0\rangle + (-1)^{1 \cdot 2} |1\rangle + (-1)^{2 \cdot 2} |2\rangle + (-1)^{3 \cdot 2} |3\rangle \right)$$

$$0 \cdot 2 = (0 \text{ AND } 1) \oplus (0 \text{ AND } 0) = 0 \oplus 0 = 0$$

$$1 \cdot 2 = (0 \text{ AND } 1) \oplus (1 \text{ AND } 0) = 0 \oplus 0 = 0$$

$$2 \cdot 2 = (1 \text{ AND } 1) \oplus (0 \text{ AND } 0) = 1 \oplus 0 = 1$$

$$3 \cdot 2 = (1 \text{ AND } 1) \oplus (1 \text{ AND } 0) = 1 \oplus 0 = 1$$

$$= H^{\otimes 2}|2\rangle = \frac{1}{2}(|0\rangle + |1\rangle - |2\rangle - |3\rangle)$$

6. $H^{\otimes 2}|2\rangle = (H|1\rangle) \otimes (H|0\rangle)$

$$= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$= \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle - |10\rangle - |11\rangle)$$

7. $H^{\otimes 2} = H \otimes H$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

We know $|2\rangle = |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$

$$H^{\otimes 2}|10\rangle = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

$$= \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle)$$

8.

$$|\psi\rangle = \alpha|100\rangle + \beta|101\rangle + \gamma|110\rangle + \delta|111\rangle$$

$$|\phi\rangle = \alpha|100\rangle + \beta|110\rangle + \gamma|101\rangle + \delta|111\rangle$$

So it doesn't have any effect even if we implement CNOT gates starting from LSB or MSB

$U_{xor, right}$, $U_{xor, center}$, $U_{xor, left}$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = U_{swap}$$

9. Given LSB as controlling bit

$$U_{xor}|00\rangle = |0 \oplus 0, 0\rangle = |0, 0\rangle = |00\rangle$$

$$U_{xor}|01\rangle = |0 \oplus 1, 1\rangle = |1, 1\rangle = |11\rangle$$

$$U_{xor}|10\rangle = |1 \oplus 0, 0\rangle = |1, 0\rangle = |10\rangle$$

$$U_{xor}|11\rangle = |1 \oplus 1, 1\rangle = |0, 1\rangle = |01\rangle$$